Assignment Module - 1

① Define data science and explain its life cycle

Definition :-

Data science is an interdisciplinary field that uses scientific methods, Algorithms, programming and statistical technique to extract useful information and insights from structural and unstructural data.

It combines concept from :-

- Statistics
- Mathematics
- computer science
- Machine learning
- Domain Knowledge

Data science life cycle :-

1. Data collection :-

Gathering data from different sources such as: databases, sensors, websites, APIs and files.

2. Data preprocessing :-

* cleaning and preparing raw data for analysis.
* Includes handling missing values, removing duplicates and correcting errors.

3. Exploratory data analysis (EDA) :-

* Analyzing data using statistical methods and visualization techniques.
* Helps to find patterns and relationships.

4. Data modeling :-
Applying machine learning algorithms to build models.

5. Model evaluation :-
checking the performance and accuracy of the model.

6. Deployment :-
Implementing the final model in a real-world applications.

7. Monitoring and maintainence :-
continuously monitoring the model and updating it when required.

② Differentiate descriptive, diagnostic, predictive, and perspective analytics ?

Type of analytics	Meaning	Question answered	Example
Descriptive analytics	Analyzes past data to understand what happened.	what will happened ?	Monthly sales report.
Diagonstic Analytics	finds the reason behind an event.	why did it happen ?	finding reasons for sales decrea
Predictive analytics	uses historical data to predict future outcomes	what will happen ?	suggesting bestest marketing strategy.

perspective analytics	suggests the actions to achieve better results.	what should be done ?	Predicting future sales and giving suggestions

summary :-

- Descriptive → past information
- Diagnostic → Reasons behind events.
- predictive → future predictions.
- perspective → Recommended actions.

③ Explain the steps involved in data preprocessing with examples.

Definition :- Data preprocessing is the process of collecting the raw data into a clean and suitable format for analysis and machine learning.

Steps involved :-

1. Data cleaning :- Removing errors, duplicating and handling missing values.
2. Data integration :- combining data from multiple pages or the sources into a single dataset.
3. Data transformation :- converting data into a suitable format.

   Includes :-

   * Normalization
   * scaling
   * Encoding
4. Data Reduction :- Reducing the size of data while maintaining important information.

5. Data Reduction / discretization :-
converting continuous data into groups of the intervals.

Conclusion :-
Data preprocessing improves data quality and helps to machine learning models produce accurate results.